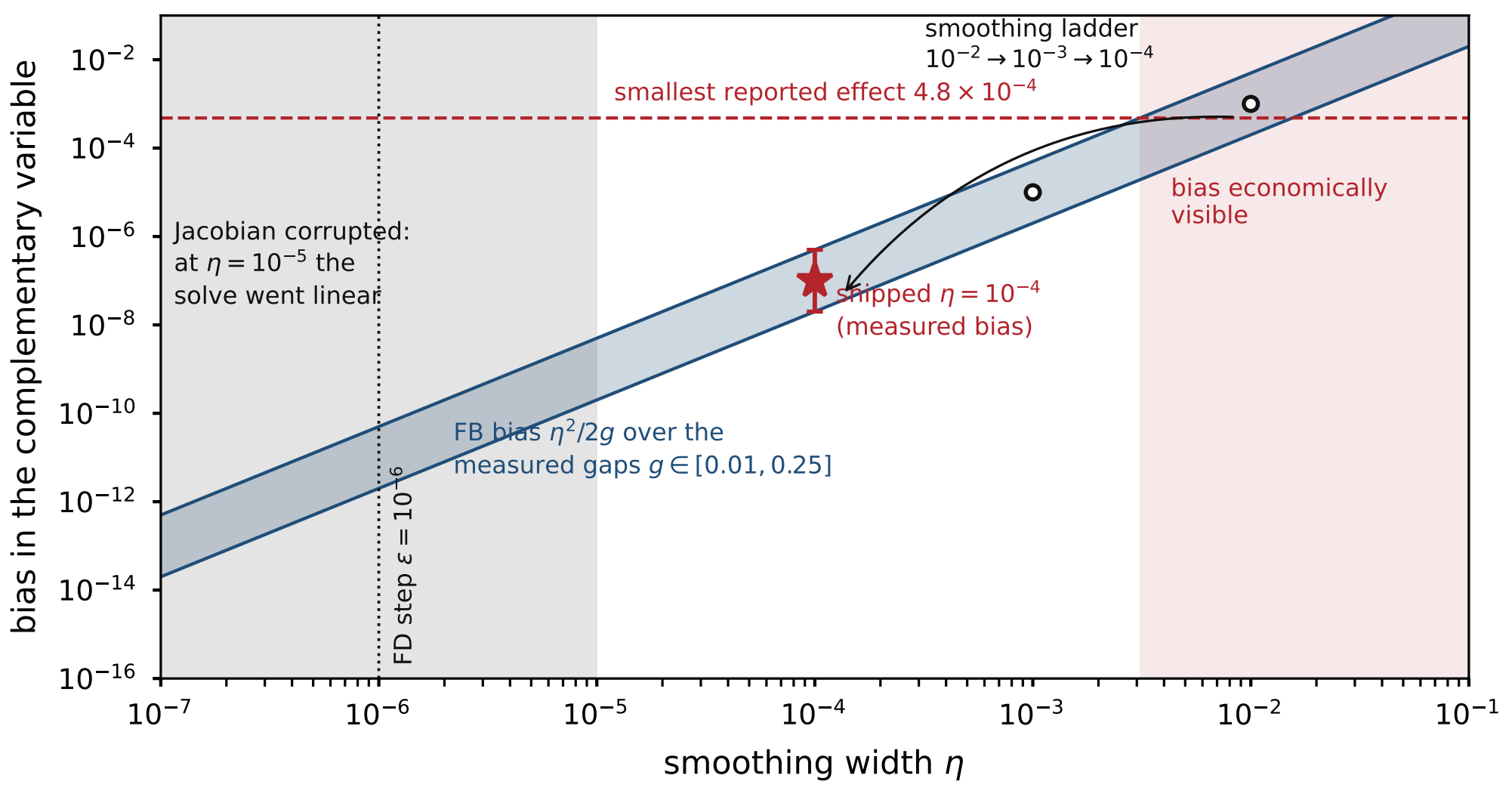
smoothing ladder
$10^{-2} \to 10^{-3} \to 10^{-4}$
smallest reported effect $4.8 \times 10^{-4}$
bias economically visible
Jacobian corrupted: at $\eta = 10^{-5}$ the solve went linear
FD step $\varepsilon = 10^{-6}$
shipped $\eta = 10^{-4}$ (measured bias)
FB bias $\eta^2/2g$ over the measured gaps $g \in [0.01, 0.25]$
bias in the complementary variable
smoothing width $\eta$
$10^{-2}$
$10^{-4}$
$10^{-6}$
$10^{-8}$
$10^{-10}$
$10^{-12}$
$10^{-14}$
$10^{-16}$
$10^{-7}$
$10^{-6}$
$10^{-5}$
$10^{-4}$
$10^{-3}$
$10^{-2}$
$10^{-1}$